Mentor Round-II Report: Two-tail Flux Statistics and Fluctuation-Theorem Diagnostics for the Boundary-driven Resonant NLS Chain

Detailed numerical report and reproducibility package

30 August 2026

Abstract

This report answers the mentor's second-round questions about the two tails of the finite-time flux distribution and whether the boundary-driven resonant NLS chain follows a fluctuation theorem. For each $n = 10, 20, 30, 40$, we generated 1,000,064 non-overlapping blocks of length 20 after burn-in and aggregated them to $\tau = 20, 40, \ldots, 200$. The central distribution is increasingly Gaussian-like, but the $\tau = 20$ far tails are heavier than the Gaussian fixed by the full-sample variance. Wherever raw tail counts remain adequate, $\log P_\tau$ is approximately linear in τ at fixed threshold, giving resolved finite-time large-deviation rate proxies. Direct medium-entropy and heat-current symmetry slopes do not reach their asymptotic references before the negative tails become unresolved, so direct sampling alone gives "not verified in the accessible window", not a violation. Two independent $n = 2$ total-entropy calculations pass finite-time detailed and integral fluctuation-relation audits. Finally, controlled cloning calculations at $n = 10, 20, 30, 40$ pass all frozen support and convergence gates and are numerically consistent with the Gallavotti–Cohen SCGF symmetry at one resolved complementary tilt pair per length. The latter is a finite-size, tested-window numerical statement, not an all-tilt proof or a thermodynamic- limit theorem.

1 The mentor's questions and the implemented answer

The requested workflow was:

1. after burn-in, store a long trajectory and form approximately 10^6 samples of the flux measured over segments of length 20;
2. describe the central 1–99% distribution by a histogram;
3. compare both tails with a normal survival function;
4. use a finite-count correction when a tail count is small;
5. plot the positive and negative probabilities and test their dependence on the threshold A and averaging time τ;
6. determine whether the fluctuation-theorem symmetry is supported.

All requested data products were generated. Two technical corrections were needed to make the test scientifically well defined. First, adding two successes and two failures gives the display-only plus-four estimator

$$\widehat{p}_{+4} = \frac{k+2}{N+4}, \tag{1}$$

not $(k+2)/(N+2)$. The denominator increases by four because four artificial observations are added in total. Raw nonzero counts, rather than this smoothed value, are used in all inferential fits. Second, the FT diagnostic is the logarithm of a probability ratio,

$$\log \frac{p_\tau(A)}{p_\tau(-A)}, \tag{2}$$

not the ratio $\log p_\tau(A)/\log p_\tau(-A)$. Equation (2) is the quantity related to entropy production.

Mentor request	Implementation	Status and interpretation
10^6 samples at $\tau = 20$	1,000,064 blocks for every n, 128 independent streams	complete and audited
Central histogram	empirical PDF over the central 1–99% region	complete; increasingly Gaussian-like
Normal tail fit	empirical survival versus $\overline{\Phi}((x-\mu)/\sigma)$ and a joint two-tail Gaussian fit	complete; descriptive, not an FT test
Small-count correction	$(k+2)/(N+4)$ for visualization only	complete; zero-count tails are not treated as evidence
Dependence on A and τ	threshold grid $0.01, 0.02, \ldots$ and $\tau = 20, 40, \ldots, 200$	complete where raw counts resolve the tail
Fluctuation theorem	direct ratio, $n = 2$ total entropy, and long-time SCGF cloning	direct test inconclusive; $n = 2$ passes; long chains are consistent at tested pairs

Table 1: One-to-one map from the mentor's request to the completed analyses.

2 Dynamics, heat, entropy, and current

All accepted samplers use projection-free Cartesian variables $c_j = x_j + iy_j$. There is no action floor or positivity projection. The bath parameters are

$$(T_L, T_R, \gamma) = (10, 2, 0.1), \qquad \Delta t = 5 \times 10^{-4}, \tag{3}$$

unless a timestep control is explicitly stated. The direct production uses burn-in $B = 500$.

Bath heat Q_r is accumulated from the energy change across the corresponding bath-only substep, with heat positive into the chain. It is not replaced by the whole-step energy change. The medium entropy is

$$\Sigma_\tau^{\mathrm{m}} = -\frac{Q_L(\tau)}{T_L} - \frac{Q_R(\tau)}{T_R}. \tag{4}$$

Every base block stores $Q_L, Q_R, \Delta E, \Sigma^{\mathrm{m}}$, the bulk action current, stream and block identifiers, and the first-law residual $Q_L + Q_R - \Delta E$.

Three questions must be kept separate:

- the distribution and large-deviation shape of the action current;
- the finite-time fluctuation relation for total thermodynamic entropy;
- the long-time Gallavotti–Cohen symmetry of the entropy-current SCGF.

The action current is not assumed to equal entropy production.

3 Direct two-tail experiment

3.1 Sampling and numerical integrity

For each $n = 10, 20, 30, 40$, 128 independent streams produced 1,000,064 non-overlapping $\tau = 20$ blocks. Consecutive base blocks within each stream were grouped without overlap to form $\tau = 40, 60, \ldots, 200$. This preserves the stream identity and avoids inventing independent samples through a moving window.

The accepted audit found exactly the requested row count, no nonfinite rows, no midpoint failures, and no ordering errors. The entropy and first-law columns recompute from the stored heat and energy columns. The RMS first-law-residual rate ranges from 9.71×10^{-6} at $n = 10$ to 3.05×10^{-6} at $n = 40$; halving the timestep reduces this residual by approximately a factor of four. Equal-temperature and bath-swapped controls give the expected zero-current and current-reversal behavior. An independent analysis audit performed 75,617 checks with zero errors.

n	blocks at $\tau = 20$	$\langle J \rangle$	heat current	$\langle \Sigma^{\mathrm{m}} \rangle / \tau$
10	1,000,064	0.39809	3.21212	1.28485
20	1,000,064	0.11979	1.40691	0.56275
30	1,000,064	0.05426	0.79595	0.31839
40	1,000,064	0.03052	0.52294	0.20918

Table 2: Production means from the joint Cartesian sampler.

As an independent transport cross-check, these means obey

$$\langle J(n) \rangle = 28.889\, n^{-1.8485}, \qquad R^2 = 0.9987, \tag{5}$$

consistent with the established canonical estimate $28.75n^{-1.850}$.

3.2 Central shape and normal-tail benchmark

For a finite-time current J_τ, the Gaussian survival benchmark is

$$\mathbb{P}(J_\tau \geq A) \approx \overline{\Phi}\left(\frac{A - \mu_\tau}{\sigma_\tau} \right). \tag{6}$$

The central distribution narrows and looks increasingly Gaussian as τ grows. This is compatible with ordinary averaging near the mean. It does not imply that the large-deviation tails are Gaussian.

At $\tau = 20$, fitting one Gaussian jointly to both empirical far tails requires a width larger than the full-sample standard deviation by factors $1.134, 1.199, 1.257$, and 1.283 for $n = 10, 20, 30, 40$. Thus the far tails are systematically heavier than the Gaussian fixed by the central variance.

3.3 Dependence on threshold and averaging time

For each threshold on the 0.01 grid, both signs were counted after sorting the samples. A fit was allowed only when every included point had at least 20 raw events, probability at most 0.2, and at least four averaging times. The fitted form was

$$\log \mathbb{P}(J_\tau \geq A) = b_+(A) - \tau I_+(A), \qquad \log \mathbb{P}(J_\tau \leq -A) = b_-(A) - \tau I_-(A). \tag{7}$$

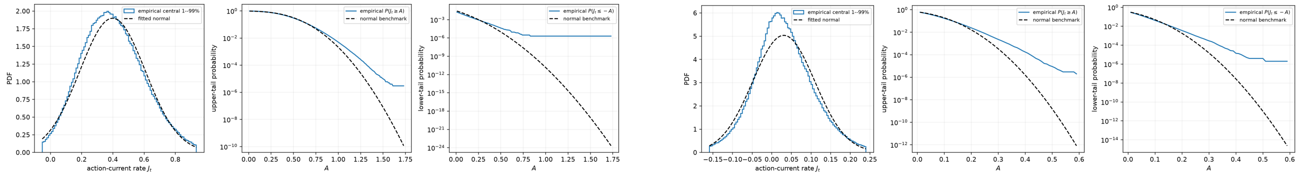

Figure 1: Representative $\tau = 20$ normal-tail comparisons. Solid empirical curves and the full-sample Gaussian benchmark disagree in the far tail; the tail-optimized Gaussian is shown only as a descriptive comparison.

Every accepted fit has $R^2 \geq 0.98$. The rate proxies $I_\pm(A)$ increase with threshold magnitude. The resolved positive-tail ranges are $[0.46, 0.81]$, $[0.15, 0.33]$, $[0.08, 0.20]$, and $[0.05, 0.15]$ for $n = 10, 20, 30, 40$. Negative-tail fits are resolved over $[0.01, 0.06]$ for $n = 20$ and $[0.01, 0.08]$ for $n = 30, 40$. At $n = 10$, the negative current becomes too rare to sustain four resolved times.

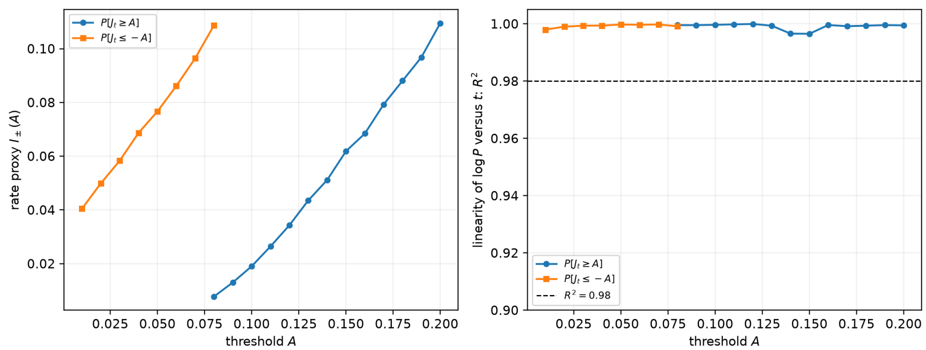

Figure 2: $n = 30$ finite-time rate proxies. Approximate linearity in τ at fixed A is evidence for finite-time large-deviation scaling; it is not by itself a fluctuation theorem.

The answer to "how does $\log P$ depend on A and τ?" is therefore: at fixed A, the resolved rare-tail log probability is approximately linear in τ; at fixed τ, it is curved in A, so one exponential in A does not describe the whole tail. In large-deviation notation,

$$\mathbb{P}(J_\tau \approx A) \asymp e^{-\tau I(A)}, \tag{8}$$

with a nonlinear rate function $I(A)$.

3.4 Direct fluctuation-symmetry result

For the medium-entropy rate $a = \Sigma_\tau^{\mathrm{m}}/\tau$, the simple asymptotic reference is

$$\frac{1}{\tau} \log \frac{p_\tau(a)}{p_\tau(-a)} = a. \tag{9}$$

The directly fitted slopes are below one. At $\tau = 20$, they are $0.653, 0.341, 0.247$, and 0.208 for $n = 10, 20, 30, 40$. Where the negative tail remains well resolved they initially move upward with τ; for example, the $n = 30$ slopes are $0.247, 0.316$, and 0.393 at $\tau = 20, 40, 60$. They do not reach one before the two-sided histogram loses support.

This is not evidence that the FT is violated. An exact finite-time relation involves the total trajectory entropy

$$\Sigma_\tau^{\mathrm{tot}} = \Sigma_\tau^{\mathrm{m}} + \log \rho(Z_0) - \log \rho(Z_\tau), \tag{10}$$

n	$N_-(\Sigma^{\mathrm{m}}; 20)$	$N_-(\Sigma^{\mathrm{m}}; 200)$	$s_{\Sigma^{\mathrm{m}}}(20)$	$s_Q(20)$	$\Delta\beta$
10	7,158	0	0.653	0.217	0.4
20	107,491	0	0.341	0.126	0.4
30	214,942	4	0.247	0.081	0.4
40	276,375	195	0.208	0.058	0.4

Table 3: Direct-sampling resolution and symmetry diagnostics. Here $N_-(X;\tau)$ counts negative blocks, while $s_{\Sigma^{\mathrm{m}}}$ and s_Q are the fitted antisymmetry slopes. Sparse later tails are not replaced by normal extrapolation or plus-four counts.

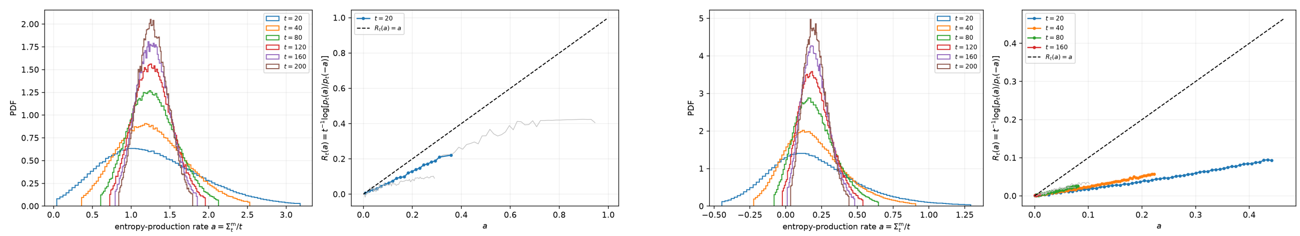

Figure 3: Representative medium-entropy symmetric-bin diagnostics. The fit uses only raw nonzero counts from both signs.

whereas the million-block long-chain data do not reconstruct the NESS endpoint density. Boundary terms and exponentially vanishing negative-tail support can therefore prevent medium entropy from displaying the asymptotic symmetry at accessible times. The correct direct-sampling conclusion is:

> The standard thermodynamic symmetry is not verified in the directly sampled finite-time window. The observed slope deficit is endpoint- and rare-tail limited and is not a demonstrated violation.

3.5 Action current is not entropy production

Heat and action become more correlated as the averaging time grows, but they are not tightly coupled. At $\tau = 200$, the Pearson correlations are $0.934, 0.896, 0.803$, and 0.683 for $n = 10, 20, 30, 40$, while the fraction of heat variance left after linear regression on action is $0.128, 0.197, 0.355$, and 0.533. Consequently, a visually attractive positive/negative action-current ratio cannot be called a thermodynamic FT, especially for longer chains.

4 Finite-time total-entropy controls at $n = 2$

The endpoint issue in (10) can be addressed at $n = 2$, where both sites are thermostatted and the stationary density can be estimated in a reduced phase-invariant coordinate system. The accepted driven calculation uses 1,048,576 non-overlapping blocks of duration 0.1 after burn-in, with an independently validated equilibrium data set used to select the density representation.

The resulting total-entropy detailed-FT fit is

$$\log \frac{p(s)}{p(-s)} = -0.02219 + (0.994496 \pm 0.00738)s. \tag{11}$$

The integral relation gives

$$\log \mathbb{E}e^{-\Sigma^{\mathrm{tot}}} = 0.002175, \qquad 95\%\ \mathrm{CI} = [-0.000744, 0.005251], \tag{12}$$

with exponential-weight ESS 324,604. Support, equilibrium-density, stationarity, sensitivity, and first-law gates pass.

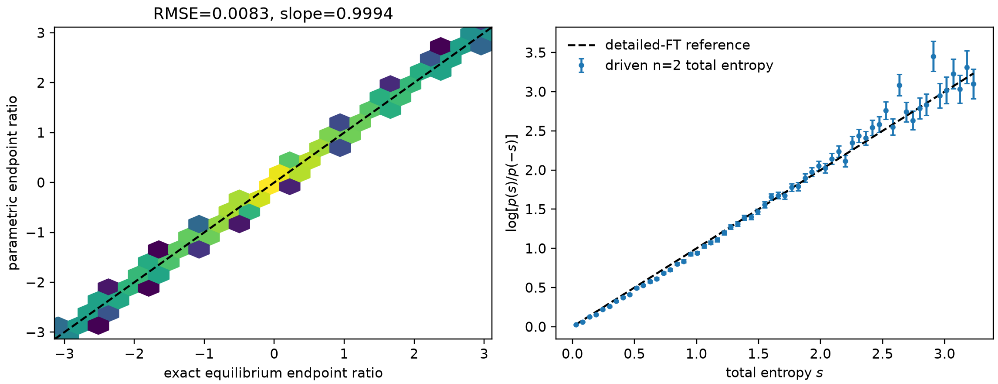

Figure 4: Accepted $n = 2$ endpoint-density validation and driven NESS total-entropy detailed relation.

An independent discrete-path calculation avoids stationary-density estimation entirely. It accumulates the exact forward/reverse Gaussian transition-kernel ratio of the split integrator for one million trajectories in each direction. Its Crooks fit is

$$\log \frac{p_F(s)}{p_R(-s)} = 0.000259 + (0.993352 \pm 0.00243)s. \tag{13}$$

Both forward and reverse integral relations pass. Timestep refinement shows first-order convergence of the exact discrete kernel entropy to the physical bath-heat entropy.

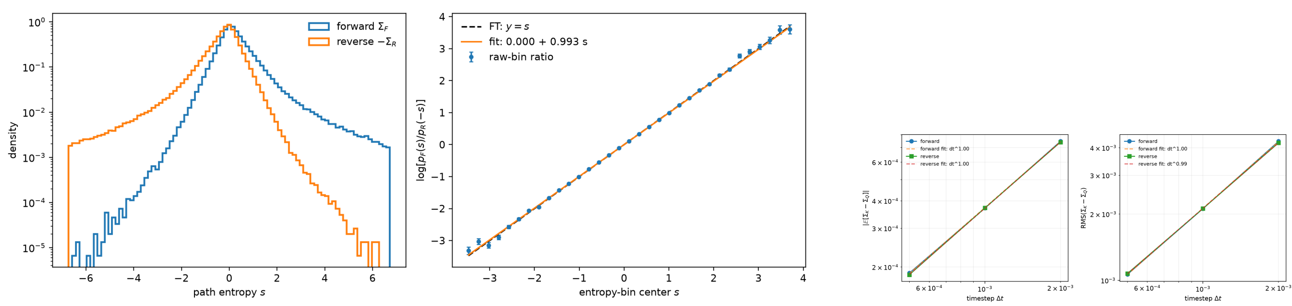

Figure 5: Independent finite-step path-ratio FT and its continuous-time heat interpretation.

Together these experiments support a paper-level finite-time numerical claim for the two-site model. They do not establish the same statement for a long chain.

5 Long-chain Gallavotti–Cohen SCGF test

Direct sampling cannot efficiently resolve exponentially rare negative tails at long times. The long-chain study therefore estimates the scaled cumulant generating function

$$\psi_n(k) = \lim_{t\to\infty} \frac{1}{t} \log \mathbb{E} e^{-k\Sigma_R(t)}, \qquad \Sigma_R = -(T_R^{-1} - T_L^{-1})Q_R = -0.4Q_R. \tag{14}$$

The Gallavotti–Cohen diagnostic is

$$\psi_n(k) = \psi_n(1-k). \tag{15}$$

The cloning protocol freezes complementary tilts, four independent seeds per member, support thresholds, observation horizons, and population, timestep, and selection-interval controls. Phase II contains 56 complete summaries, 56 timeseries, and 56 logs. No output is malformed or nonfinite, no midpoint solve fails, and all prospective gates pass.

n	$(k, 1-k)$	clones	horizon	$\psi(k) - \psi(1-k)$	95% CI	gates
10	$(0.3, 0.7)$	2048	60	-0.000075	$[-0.009953,\ 0.009802]$	pass
20	$(0.4, 0.6)$	1024	60	-0.003389	$[-0.012081,\ 0.005302]$	pass
30	$(0.4, 0.6)$	4096	60	-0.002336	$[-0.007251,\ 0.002578]$	pass
40	$(0.4, 0.6)$	1024	120	$+0.004498$	$[-0.002343,\ 0.011340]$	pass

Table 4: Final long-chain paired SCGF residuals. "Pass" includes support, time, population, timestep, and selection controls applicable to that row.

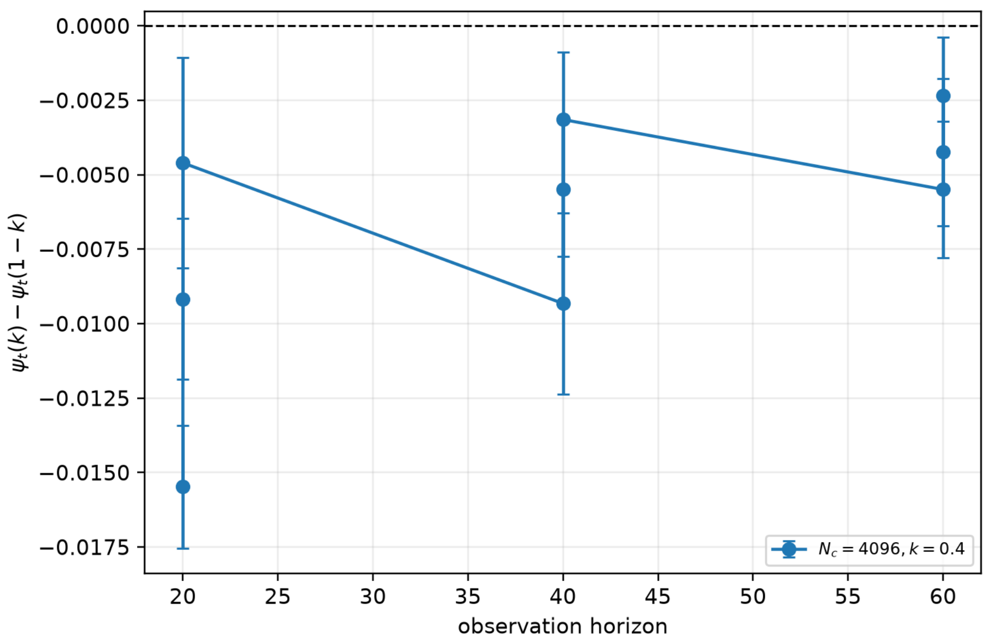

Figure 6: $n = 30$ observation-time, population, timestep, and selection convergence of the paired SCGF residual.

Every confidence interval in Table 4 contains zero and every absolute residual is below the frozen tolerance. The normalized residuals are approximately $-0.04\%, -4.0\%, -4.9\%$, and 12.9%. The $n = 40$ absolute gate passes, but its relative uncertainty is larger because the SCGF magnitude is smaller. This caveat must remain visible.

The admissible long-chain claim is:

> For the boundary-driven projection-free Cartesian NLS chain at $(T_L, T_R, \gamma) = (10, 2, 0.1)$, controlled rare-event estimates are consistent with Gallavotti–Cohen SCGF symmetry at all four tested finite chain lengths over the resolved complementary tilt pair.

It is not legitimate to claim exact equality, symmetry over all $k \in [0, 1]$, uniformity in n, or an $n \to \infty$ theorem from these data.

6 Final answer to the second-round questions

1. **Was the requested sampling completed?** Yes. The requested million-sample scale was reached for all four chain lengths, with complete heat, entropy, current, endpoint-audit, and stream metadata.
2. **What is the flux distribution?** Its centre becomes Gaussian-like under longer averaging, while the $\tau = 20$ far tails are heavier than the full-sample Gaussian benchmark.
3. **How does $\log P$ depend on A and τ?** At fixed resolved A, it is approximately affine in τ; the fitted decay rate grows with $|A|$. At fixed τ, the dependence on A is curved, consistent with a nonlinear finite-time rate function.
4. **Does the directly sampled current obey the FT?** The direct medium-entropy and heat ratios do not reach their references in the accessible window. This is an unresolved finite-time test, not a demonstrated failure. Action-current symmetry has no thermodynamic target.
5. **Can the FT be verified numerically at all?** Yes, with scoped observables. At $n = 2$, total entropy passes two independent finite-time checks. At $n = 10, 20, 30, 40$, long-time entropy-current cloning is consistent with the GC SCGF symmetry at the tested pairs after all frozen controls.

7 Code, data, and provenance map

Module	Principal code	Accepted results
Direct tails	`code/direct_sampling/NLS_entropy_ft.cpp`, `analyze_entropy_ft.py`, `plot_entropy_ft_supplement.py`, `analyze_action_tail_time_scaling.py`	`results/direct_sampling/`
$n = 2$ endpoint entropy	`code/n2_total_entropy/analyze_total_entropy_parametric_n2.py` and independent audit	`results/n2_total_entropy/`
Discrete path ratio	`code/discrete_path/NLS_discrete_path_ft.cpp` and `analyze_discrete_path_ft.py`	`results/discrete_path/`
Long-chain SCGF	`code/long_chain_scgf/NLS_entropy_cloning.cpp`, `analyze_long_chain_ft.py`, `analyze_long_chain_ft_controls.py`	`results/long_chain_phase2/`

Table 5: Traceability from each scientific claim to code and accepted data.

The compact package contains all code, result tables, audits, figures, protocols, and all Phase-II raw summaries/timeseries. The multi-gigabyte block and trajectory CSVs are kept once in the main local repository. Their absolute locations, byte counts, and SHA-256 hashes are listed in `RAW_DATA_INDEX.md`. Reproduction commands are in `REPRODUCE.md`; the package-level integrity list is `SHA256SUMS.txt`.

Claim boundary for manuscript use

The direct distribution study can be reported as finite-time current fluctuations and large-deviation evidence. The $n = 2$ result can be reported as a controlled finite-time total-entropy numerical verification. The long-chain result can be reported as controlled numerical consistency with GC symmetry at the tested complementary pairs. None of these numerical results alone proves the continuous-time theorem or the infinite-chain limit.